Week 9 Session A

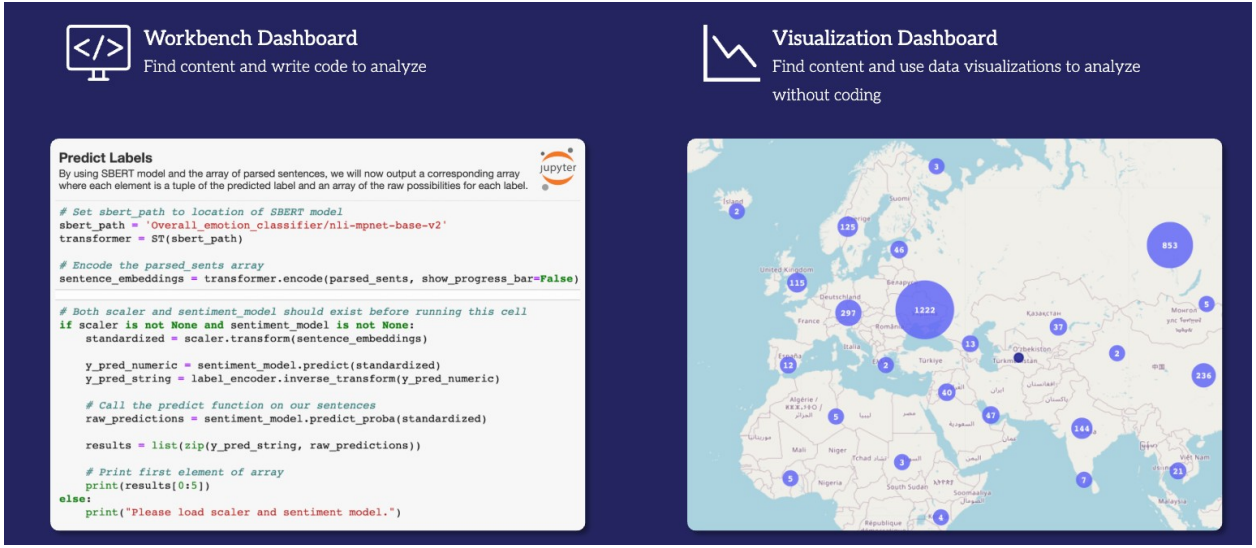
Tuesday, March 17 2026

Overview

- ProQuestTDM (text and data mining)

Text Analysis and ProQuestTDM

- **Dataset:**
  - Search terms: “data center” AND Michigan
  - Timeframe: 1/1/2025 - 3/10/2026
  - Sources: US Newsstream Collections, newspapers
  - Results: 469 documents



Text Analysis and ProQuestTDM

- **Visualization Dashboard on Topic Modeling**
  - When you clicked Topic Modeling, what are your findings?
    - How many topics did you review?
    - What do you like about this dashboard?
    - What are some potential limitations?
    - Let’s compare our results!

Text Analysis and ProQuestTDM

- **Workbench in ProQuestTDM: Hands-on coding!**
  - Let's run things in Jupyter Notebook and see the backend of this!
- What does the data actually look like?
  - .xml files


```xml
<?xml version="1.0" encoding="UTF-8"?><RECORD>
   <GOID>3152822073</GOID>

   <Obj>
      <SourceRollupType>Newspapers</SourceRollupType>
      <ObjectTypes>
         <other ObjectTypeOrigin="EXT">News</other>
         <mstar>News</mstar>
      </ObjectTypes>
      <ObjectRollupType>Articles - All Types</ObjectRollupType>
      <TitleAtt HTMLContent="true">
         <Title>Trump: Michigan set to receive part of $20B in new data centers</Title>
      </TitleAtt>
      <NumericDate>2025-01-09</NumericDate>
      <StartDate>2025-01-09</StartDate>
      <EndDate>2025-01-09</EndDate>
      <AlphaDate>Jan 9, 2025</AlphaDate>
      <Language IsPrimary="true">
         <ISO>
            <ISOCode>ENG</ISOCode>
            <ISOExpansion ISOCode="ENG">English</ISOExpansion>
         </ISO>
      </Language>
      <Language>
         <RawLang>English</RawLang>
      </Language>
```

Our results vs ProQuestTDM’s results

- Are they the same?
- If not, why?
  - LDA is a probabilistic model, so small differences between runs are normal, but when we fix the random seed and keep preprocessing constant, the results stabilize. ProQuest’s topics look consistent because they run the model once with fixed settings and reuse the results.

Week 9 Assignment

- Examine the results generated from ProQuest's built in function of topic modeling.